Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 9 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第9章 · 积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C09-R01 · P1 · Chapter 9

The curve C has gradient function $\frac{dy}{dx} = 6x^2 - 8x + 3$ and passes through $(1, 5)$.

(a) Find an equation of C. (4)

(b) Find the equation of the tangent at $(1, 5)$. (3)

(Total for Question 1 is 7 marks)

For $x > 0$, $\dfrac{dy}{dx} = 3\sqrt{x} - \dfrac{4}{x^2}$. The curve passes through $(4, 7)$.

(a) Find y in terms of x. (4)

(b) Find the point on the curve with $x = 1$. (2)

(Total for Question 2 is 6 marks)

A curve has $\frac{d^2y}{dx^2} = 12x - 6$. At $x = 1$, the curve has gradient 4 and passes through $(1, 2)$.

(a) Find dy/dx. (3)

(b) Find the equation of the curve. (3)

(Total for Question 3 is 6 marks)

Find each indefinite integral, simplifying your answer.

(a) $\int (8x^3 - 6x + 5)\, dx.$ (3)

(b) $\int \dfrac{6x^4 - 4x}{x^3}\, dx$, for $x > 0$. (2)

(Total for Question 4 is 5 marks)

5. P1-C09-R05 · P1 · Chapter 9

The derivative of $f(x)$ is $4x^3 + 2x$. It is given that $f(0) = f(1) - 2$.

(a) Show that this condition does not determine the integration constant. (3)

(b) Given also that $f(2) = 23$, find $f(x)$. (2)

(Total for Question 5 is 5 marks)

6.

The curve C has $dy/dx = (x - 1)(3x + 2)$ and passes through the origin.

(a) Find an equation of C. (4)

(b) Find the exact coordinates of the points where C has a horizontal tangent. (3)

(Total for Question 6 is 7 marks)

For $x > 0$, $dy/dx = 5x^{3/2} - 3x^{-1/2}$. At $x = 1$, the tangent to the curve is $y = 2x + 1$.

(a) Find the coordinates of the point of contact. (2)

(b) Find the equation of the curve. (4)

(Total for Question 7 is 6 marks)

8. P1-C09-R08 · P1 · Chapter 9

A curve has $dy/dx = ax + b$. Its gradient is 1 at $x = 0$ and 7 at $x = 3$. It passes through $(2, 9)$.

(a) Find a, b. (3)

(b) Find an equation of the curve and its y-intercept. (4)

(Total for Question 8 is 7 marks)

9. P1-C09-R09 · P1 · Chapter 9

The curve C has $dy/dx = 4x - 6$ and passes through $(3, 5)$.

(a) Find an equation of C. (3)

(b) Find the normal at $(3, 5)$ and where it meets the y-axis. (3)

(Total for Question 9 is 6 marks)

10. P1-C09-R10 · P1 · Chapter 9

Two curves have the same gradient function $dy/dx = 3x^2 - 2$. One passes through $(0, 4)$ and the other through $(2, 1)$.

(a) Find equations of both curves. (4)

(b) Describe the single translation mapping C_1 onto C_2. (2)

(Total for Question 10 is 6 marks)

Find the following integrals for $x > 0$.

(a) $\displaystyle\int \frac{9x^2 + 3}{\sqrt{x}}\,dx.$ (4)

(b) $\displaystyle\int (\sqrt{x} - 2)^2\,dx.$ (4)

(Total for Question 11 is 8 marks)

The curve C satisfies $dy/dx = 6x^2 + k$. It passes through $(0, 1)$ and $(1, 6)$.

(a) Find k and the equation of C. (4)

(b) Find the tangent to C at $(1, 6)$. (3)

(Total for Question 12 is 7 marks)

13.

The gradient of a curve is $dy/dx = 12x^2 - 12x$. The curve has a horizontal tangent at a point with coordinates $(1, 3)$.

(a) Find the equation of the curve. (3)

(b) Find the other point where the tangent is horizontal and state the equations of both horizontal tangents. (3)

(Total for Question 13 is 6 marks)

For $x > 0$, $dy/dx = \dfrac{2x^2 - 3}{x^2}$. The curve passes through $(3, 4)$.

(a) Find y. (4)

(b) Find all intersections of the curve with $y = 2$. (3)

(Total for Question 14 is 7 marks)

15.

A curve has $d^2y/dx^2 = 6x + 4$. Its tangent at the origin is $y = -2x$.

(a) Find dy/dx. (3)

(b) Find the equation of the curve. (3)

(Total for Question 15 is 6 marks)

$F(x) = ax^4 + bx^2 + c$ is an antiderivative of $12x^3 - 10x$. It is given that $F(1) = 0$.

(a) Find a, b, c. (4)

(b) Factorise $F(x)$ completely into factors with integer coefficients. (3)

(Total for Question 16 is 7 marks)

17.

The gradient of C at any x is $8x + 6$. At the point where $x = 1$, the normal passes through $(0, 5)$.

(a) Find the coordinates of the point on C with $x = 1$. (3)

(b) Find the equation of C. (3)

(Total for Question 17 is 6 marks)

18.

For $x > 0$, $dy/dx = \dfrac{5}{\sqrt{x}} + \dfrac{6}{x^3}$. It is given that $y = 17$ when $x = 1$.

(a) Find y in terms of x. (4)

(b) Find y when $x = 4$, giving an exact answer. (2)

(Total for Question 18 is 6 marks)

19.

A curve has $dy/dx = 2ax + b$. It passes through $(0, 2)$ and $(2, 10)$ and has gradient 3 at $x = 1$.

(a) Show that no such curve exists. (4)

(b) If the stated gradient is changed from 3 to 4, determine whether a is uniquely fixed. Give a reason. (2)

(Total for Question 19 is 6 marks)

The curve C has $dy/dx = 3x^2 - 6x + 2$ and passes through $(2, 1)$.

(a) Find the equation of C. (3)

(b) Find the equation of the tangent at the y-intercept of C. (3)

(Total for Question 20 is 6 marks)

For $x > 0$, $dy/dx = \frac{3}{2}x^{1/2} + 4$. The curve passes through $(4, 25)$.

(a) Find y. (3)

(b) Find the tangent at $x = 4$ and where this tangent meets the y-axis. (3)

(Total for Question 21 is 6 marks)

The curve C has $d^2y/dx^2 = 24x^2 - 4$. It has a horizontal tangent at $(1, 2)$.

(a) Find dy/dx. (3)

(b) Find the equation of C. (3)

(Total for Question 22 is 6 marks)

The derivative of $f(x)$ is $(2x-1)^2$.

(a) Find $f(x)$ in terms of an arbitrary constant. (4)

(b) Given that $f(0) = 2$, find

(i) $f(\frac{1}{2})$, (2)

(ii) the gradient of $y = f(x)$ when $x = \frac{1}{2}$. (2)

(Total for Question 23 is 8 marks)

The curve C has $dy/dx = 5x^4 - 4x^3$. Its y-intercept is -2.

(a) Find an equation of C. (3)

(b) Find the coordinates of the two points where $dy/dx = 0$, and state the equation of the tangent at $x = 0$. (4)

(Total for Question 24 is 7 marks)

A curve has $dy/dx = 2x - \dfrac{6}{x^2}$ for $x > 0$. Its tangent at $x = 1$ passes through $(0, 10)$.

(a) Find the coordinates of the point on the curve with $x = 1$. (3)

(b) Find an equation of the curve. (3)

(Total for Question 25 is 6 marks)

For a curve C, $dy/dx = 6x^2 + ax$ and $y(0) = 3$. At $x = 2$, the tangent is parallel to $y = 16x + 1$.

(a) Find a. (2)

(b) Find the equation of C and the equation of its tangent at $x = 2$. (4)

(Total for Question 26 is 6 marks)

Find the following indefinite integrals for $x > 0$.

(a) $\displaystyle\int x^{1/3}(x+2)\,dx.$ (4)

(b) $\displaystyle\int \frac{x^4-1}{x^2}\,dx.$ (4)

(Total for Question 27 is 8 marks)

28. P1-C09-R28 · P1 · Chapter 9

The gradient function of C is $dy/dx = 4x^3 - 6x^2$. The curve meets the x-axis at $x = 2$.

(a) Find the equation of C. (3)

(b) Find all x-intercepts of C and state which also have a horizontal tangent. (3)

(Total for Question 28 is 6 marks)

29. P1-C09-R29 · P1 · Chapter 9

A curve has $d^2y/dx^2 = 6x$. Its normal at $(1, 3)$ has gradient $-\frac{1}{2}$.

(a) Find dy/dx. (3)

(b) Find the equation of the curve and verify that it passes through $(0, 3)$. (4)

(Total for Question 29 is 7 marks)

For $x > 0$, $dy/dx = \dfrac{3x^2 + 2x + 1}{x^{3/2}}$. The curve passes through $(1, 0)$.

(a) Find y in terms of x. (4)

(b) Find the equation of the normal at $(1, 0)$. (3)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C09-R01 · 7 marks

(a) · 4 marks

$y = 2x^3 - 4x^2 + 3x + 4.$

逐项积分得$2x^3-4x^2+3x+C$；代$(1,5)$得$C=4$。

M1	正确积分三项
M1	加常数并代点
A1	积分常数
A1	完整方程

(b) · 3 marks

$y = x + 4.$

由给定梯度函数在1处得1；过$(1,5)$写点斜式。

M1	从梯度式求斜率
M1	点斜式
A1	方程

Question 2

P1-C09-R02 · 6 marks

(a) · 4 marks

$y = 2x^{3/2} + 4x^{-1} - 10$.

将根式和分式写成幂，积分得2x^{3/2}+4/x+C，代4得7=16+1+C。

M1	分数负指数积分
M1	常数条件
A1	常数负10
A1	方程

(b) · 2 marks

$(1, -4)$.

代入已确定常数的曲线得2+4−10=−4。

M1	使用完整曲线而非梯度
A1	坐标

Question 3

P1-C09-R03 · 6 marks

(a) · 3 marks

$6x^2 - 6x + 4$.

第一次积分加常数A；在1处6−6+A=4，所以A=4。

M1	一次积分
M1	用梯度条件
A1	完整一阶导

(b) · 3 marks

$y = 2x^3 - 3x^2 + 4x - 1$.

再积分并加另一个常数B，代(1,2)给2=2−3+4+B。

M1	二次积分
M1	新增独立常数及点条件
A1	方程

Question 4

P1-C09-R04 · 5 marks

(a) · 3 marks

$2x^4 - 3x^2 + 5x + C$.

逐项加1指数，再除以新指数，常数5积分成5x。

M1	幂积分
A1	三项
B1	积分常数

(b) · 2 marks

$3x^2 + 4x^{-1} + C$.

先化简为6x−4x^{-2}，再用幂积分；不含指数−1的被积项。

M1	化简
A1	积分式

Question 5

P1-C09-R05 · 5 marks

(a) · 3 marks

$f(x) = x^4 + x^2 + C$, so $f(1) - f(0) = 2$ for every C.

积分后代入两个点，C在差中抵消，给定条件对任意C都成立。

M1	正确积分含C
M1	代入两点并相减
A1	解释不确定性

(b) · 2 marks

$f(x) = x^4 + x^2 + 3$.

23=16+4+C，故C=3。

M1	使用新增绝对值条件
A1	常数与方程

Question 6

P1-C09-R06 · 7 marks

(a) · 4 marks

$y = x^3 - \frac{1}{2}x^2 - 2x.$

先展开导式$3x^2-x-2$，积分得$x^3-x^2/2-2x+C$；过原点给$C=0$。

M1	正确展开
M1	逐项积分
M1	原点条件
A1	方程

(b) · 3 marks

$(1, -\frac{3}{2})$ and $(-\frac{2}{3}, \frac{22}{27})$.

由已给因式得$x=1$或$-2/3$；代入曲线求y。

M1	梯度零点
M1	正确回代
A1	两坐标

Question 7

P1-C09-R07 · 6 marks

(a) · 2 marks

$(1, 3)$.

切点在给定切线上，将x=1代入其方程。

M1	正确使用切线条件
A1	点

(b) · 4 marks

$y = 2x^{5/2} - 6x^{1/2} + 7$.

积分后在1处y=−4+C=3，故C=7。

M1	正确两项积分
M1	用切点求常数
A1	C
A1	曲线

Question 8

P1-C09-R08 · 7 marks

(a) · 3 marks

$a = 2, b = 1.$

第一梯度直接给b=1；第二给3a+b=7。

B1	b
M1	第二梯度条件
A1	a

(b) · 4 marks

$y = x^2 + x + 3;\ (0, 3).$

积分2x+1得x^2+x+C，代(2,9)得C=3。

M1	积分
M1	点条件
A1	曲线
B1	截距

Question 9

P1-C09-R09 · 6 marks

(a) · 3 marks

$y = 2x^2 - 6x + 5$.

积分$2x^2-6x+C$，代3时前两项抵消，C=5。

M1	积分含常数
M1	代点
A1	曲线

(b) · 3 marks

$y = -x/6 + 11/2$; intercept $(0, 11/2)$.

梯度$4\times3-6=6$，法线斜率$-1/6$，过$(3,5)$。

M1	求切线与法线斜率
M1	点斜式
A1	方程与截距

Question 10

P1-C09-R10 · 6 marks

(a) · 4 marks

$C_1: y = x^3 - 2x + 4;\ C_2: y = x^3 - 2x - 3.$

共同原函数x^3-2x，各自常数由两点独立求得4和-3。

M1	正确共同原函数
M1	第一常数
M1	第二常数
A1	两曲线

(b) · 2 marks

$\begin{pmatrix}0\\-7\end{pmatrix}$.

同一x下纵坐标差恒为-7，只需下移7。

M1	比较常数差
A1	平移向量

Question 11

P1-C09-R11 · 8 marks

(a) · 4 marks

$\frac{18}{5}x^{5/2} + 6x^{1/2} + C.$

先化成9x^{3/2}+3x^{-1/2}，分别积分。

M1	两项幂化简
M1	分数幂积分
A1	两项系数
B1	常数

(b) · 4 marks

$\frac{1}{2}x^2 - \frac{8}{3}x^{3/2} + 4x + C.$

先展开成x−4√x+4，再积分，不能把整体指数直接加1。

M1	正确展开
M1	逐项积分
A1	三项
B1	常数

Question 12

P1-C09-R12 · 7 marks

(a) · 4 marks

$k = 3;\ y = 2x^3 + 3x + 1.$

积分$2x^3+kx+C$；首点给$C=1$，第二点给$6=2+k+1$。

M1	正确积分含两个未知常数
M1	第一条件
M1	第二条件
A1	k及曲线

(b) · 3 marks

$y = 9x - 3.$

梯度$6+3=9$，点斜式$y-6=9(x-1)$。

M1	梯度代值
M1	过点
A1	方程

Question 13

P1-C09-R13 · 6 marks

(a) · 3 marks

$y = 4x^3 - 6x^2 + 5$.

积分加C，在(1,3)处4−6+C=3给C=5；已给点的横坐标确使导数为0。

M1	积分
M1	使用点条件
A1	曲线

(b) · 3 marks

Other point $(0, 5)$; tangents $y = 3$ and $y = 5$.

给定导式=12x(x−1)，另一根0，曲线y(0)=5。

M1	找另一梯度零点
A1	另一点
A1	两水平线

Question 14

P1-C09-R14 · 7 marks

(a) · 4 marks

$y = 2x + 3x^{-1} - 3$.

导式先化为2−3x^{−2}，积分2x+3/x+C；在3处y=7+C=4。

M1	正确幂化简
M1	积分
M1	点条件
A1	方程

(b) · 3 marks

$(\frac{3}{2}, 2)$ and $(1, 2)$.

2x+3/x−3=2给$2x^2-5x+3=0$，因式(2x−3)(x−1)，两根均正。

M1	化为二次方程
M1	解并检查定义域
A1	两交点

Question 15

P1-C09-R15 · 6 marks

(a) · 3 marks

$3x^2 + 4x - 2.$

积分得$3x^2+4x+A$；原点切线斜率-2给$A=-2$。

M1	积分二阶导
M1	用斜率初值
A1	一阶导

(b) · 3 marks

$y = x^3 + 2x^2 - 2x.$

再次积分得x^3+2x^2-2x+B；曲线过原点给$B=0$。

M1	再积分
M1	单独确定第二常数
A1	方程

Question 16

P1-C09-R16 · 7 marks

(a) · 4 marks

$a=3, b=-5, c=2.$

积分对应a=3,b=−5；F(1)=3−5+c=0给c=2。

M1	对应四次项
M1	对应二次项
M1	点条件
A1	三常数

(b) · 3 marks

$F(x)=(3x^2-2)(x-1)(x+1).$

先作关于x^2的二次式，$3x^4-5x^2+2=(3x^2-2)(x^2-1)$，再平方差。

M1	关于x平方分解
M1	分解平方差
A1	完全因子

Question 17

P1-C09-R17 · 6 marks

(a) · 3 marks

$\left(1, \frac{69}{14}\right)$.

切线斜率14，法线−1/14；通过(0,5)故法线y=5−x/14，x=1时69/14。

M1	求法线斜率
M1	用已知法线点
A1	曲线上点

(b) · 3 marks

$y = 4x^2 + 6x - \frac{71}{14}$.

积分$4x^2$+6x+C；在1处10+C=69/14给C=−71/14。

M1	积分
M1	使用(a)点
A1	曲线

Question 18

P1-C09-R18 · 6 marks

(a) · 4 marks

$y = 10\sqrt{x} - 3x^{-2} + 10.$

积分5x^{-1/2}得10√x；积分6x^{-3}得−3x^{-2}；初值给C=10。

M1	第一项积分
M1	第二项积分
M1	确定常数
A1	完整式

(b) · 2 marks

$\frac{477}{16}$.

y=20−3/16+10=477/16。

M1	代入完整原函数
A1	精确值

Question 19

P1-C09-R19 · 6 marks

(a) · 4 marks

Point conditions imply $2a + b = 4$, but the gradient condition requires $2a + b = 3$.

积分y=ax²+bx+C，首点C=2；第二点给4a+2b=8，即2a+b=4，与导数在1处3矛盾。

M1	积分并用首点
M1	第二点约束
M1	梯度约束
A1	明确矛盾

(b) · 2 marks

No; any a with $b = 4 - 2a$ satisfies the conditions.

改后的梯度条件与两点推出的条件完全相同，只有一条独立参数关系。

M1	比较两条件是否独立
A1	正确解释并给参数关系

Question 20

P1-C09-R20 · 6 marks

(a) · 3 marks

$y = x^3 - 3x^2 + 2x + 1.$

积分各项，x=2时不含常数部分为0，故C=1。

M1	正确积分
M1	代入条件
A1	方程

(b) · 3 marks

$y = 2x + 1.$

y截距(0,1)，梯度在0处为2。

B1	截距点
M1	梯度
A1	方程

Question 21

P1-C09-R21 · 6 marks

(a) · 3 marks

$y = x^{3/2} + 4x + 1$.

积分得x^{3/2}+4x+C；8+16+C=25，所以C=1。

M1	分数指数积分
M1	初值条件
A1	曲线

(b) · 3 marks

$y = 7x - 3$; $(0, -3)$.

梯度3+4=7，过(4,25)截距25−28=−3。

M1	正确梯度
M1	点斜式
A1	切线与截距

Question 22

P1-C09-R22 · 6 marks

(a) · 3 marks

$8x^3 - 4x - 4.$

第一次积分$8x^3-4x+A$，水平切线使$8-4+A=0$。

M1	一次积分
M1	水平梯度条件
A1	一阶导

(b) · 3 marks

$y = 2x^4 - 2x^2 - 4x + 6.$

再积分$2x^4-2x^2-4x+B$，代点$1,2$给$B=6$。

M1	二次积分
M1	点条件
A1	完整式

Question 23

P1-C09-R23 · 8 marks

(a) · 4 marks

$f(x) = \frac{4}{3}x^3 - 2x^2 + x + C.$

先展开$4x^2-4x+1$，再逐项积分。

M1	正确展开
M1	幂积分
A1	三项
B1	任意常数

(b) Given that $f(0) = 2$, find

(i) · 2 marks

$\frac{13}{6}$.

$C=2$；代$1/2$得$1/6-1/2+1/2+2=13/6$。

M1	使用常数及分数代入
A1	精确值

(ii) · 2 marks

0.

直接把$1/2$代入已给导式$(2x-1)^2$。

M1	区分函数值和导数值
A1	梯度

Question 24

P1-C09-R24 · 7 marks

(a) · 3 marks

$y = x^5 - x^4 - 2$.

逐项积分，截距直接给常数−2。

M1	正确积分
M1	截距条件
A1	曲线

(b) · 4 marks

$(0, -2)$ and $(\frac{4}{5}, -\frac{6506}{3125})$; tangent $y = -2$.

导式$x^3(5x-4)=0$；x=4/5时代值为1024/3125−256/625−2=−6506/3125。

M1	找全部导数零点
M1	精确代回
A1	两点
B1	水平切线

Question 25

P1-C09-R25 · 6 marks

(a) · 3 marks

$(1, 6)$.

切线斜率-4，且截距10，故切线$y=-4x+10$，代1得6。

M1	求切线斜率
M1	用所给截距
A1	点

(b) · 3 marks

$y = x^2 + 6/x - 1$.

积分导式得$x^2+6/x+C$，代$(1,6)$求$C=-1$。

M1	负幂积分
M1	使用切点
A1	曲线

Question 26

P1-C09-R26 · 6 marks

(a) · 2 marks

$a = -4$.

导数在2处24+2a=16，所以a=−4。

M1	正确梯度条件
A1	a

(b) · 4 marks

$C: y = 2x^3 - 2x^2 + 3$; tangent $y = 16x - 21$.

积分$6x^2-4x$后常数由y(0)=3得3；x=2时y=11，切线截距11−32=−21。

M1	积分及常数
A1	曲线
M1	求切点并过点
A1	切线

Question 27

P1-C09-R27 · 8 marks

(a) · 4 marks

$\frac{3}{7}x^{7/3}+\frac{3}{2}x^{4/3}+C.$

展开成x^{4/3}+2x^{1/3}，指数各加1并除以新指数。

M1	正确幂展开
M1	两项积分
A1	系数
B1	常数

(b) · 4 marks

$\frac{1}{3}x^{3}+x^{-1}+C.$

先化简x^2-x^{-2}，第二项积分符号变为正。

M1	正确分式化简
M1	负幂积分
A1	两项
B1	常数

Question 28

P1-C09-R28 · 6 marks

(a) · 3 marks

$y = x^4 - 2x^3$.

积分x^4-2x^3+C，代$(2,0)$得$C=0$。

M1	积分
M1	横截距条件
A1	曲线

(b) · 3 marks

Intercepts $(0,0), (2,0)$; only $(0,0)$ has horizontal tangent.

曲线$x^3(x-2)=0$；导数在0处0，在2处$32-24=8$，不为0。

M1	分解求全部截距
M1	检验导数
A1	正确分类

Question 29

P1-C09-R29 · 7 marks

(a) · 3 marks

$3x^2 - 1$.

法线−1/2对应切线斜率2；第一次积分$3x^2$+A，在1处3+A=2。

M1	法切斜率关系
M1	积分及梯度条件
A1	一阶导

(b) · 4 marks

$y = x^3 - x + 3$; at $x = 0$, $y = 3$.

再积分x^3−x+B，用(1,3)得B=3；代0复核。

M1	再积分
M1	点条件
A1	曲线
B1	验证

Question 30

P1-C09-R30 · 7 marks

(a) · 4 marks

$y = 2x^{3/2} + 4x^{1/2} - 2x^{-1/2} - 4.$

导式为3x^{1/2}+2x^{−1/2}+x^{−3/2}；积分后代1得2+4−2+C=0。

M1	三项幂化简
M1	三项积分
M1	初值
A1	完整曲线

(b) · 3 marks

$y = -\frac{1}{6}x + \frac{1}{6}.$

给定导式在1处为6，所以法线斜率−1/6。

M1	梯度及负倒数
M1	过点
A1	法线